

TANISHK YADAV

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EDUCATION

New York University, Tandon School of Engineering

Master of Science in Financial Engineering

Coursework: Econometrics & Time Series, Volatility Models, Derivative Securities, Fixed Income Quant Trading, Risk Management

Brooklyn, NY

May 2027

SRM University-AP

Bachelor of Technology in Computer Science & Engineering (AI & Machine Learning); 100% merit scholarship

Coursework: Linear Algebra, Multivariable Calculus, Differential Equations, Probability & Statistics, Algorithms, Machine Learning

Amaravati, India

Jun 2025

MASTER'S THESIS

The U.S. Sovereign Debt Doom Loop: A Point-in-Time Framework for Identification, Market Pricing, and Policy Response

MS thesis, NYU Tandon; advisor Prof. David Shimko · [Overview](#)

In progress, May 2027

- **Identification.** Tests in real time when US debt dynamics turn self-reinforcing: rebuilds marketable Treasury debt bond by bond from 11,000+ auctions since 1979 and reads 39 macro series from full revision histories, so no judgment sees unreleased data
- **Market pricing.** Reads fiscal stress through the rates market, measuring the Treasury volatility risk premium (implied versus realized) and a regime-switching stress state, and tests whether each leads the fiscal gap under multiple-testing control
- **Policy response.** Simulates 30-year debt paths under two independently estimated engines, a vector autoregression and a regime-switching hidden Markov model, on common random numbers, pricing five policies against four shocks
- **Rigor.** Every assumption is tested out of sample or stated as a literature range, then challenged by an LLM agent over a purpose-built 147-paper retrieval corpus (hybrid vector and keyword search, citation graph); invariance tests bar lookahead

RESEARCH & PROJECTS

Concentrated Sectors Transmit Less [Paper](#) · [Code](#)

Submitted to Studies in Nonlinear Dynamics & Econometrics, 2026

- Linked the rise of mega-caps (top 10 firms went from 24% to 41% of the S&P 500) to how shocks spread across the 11 sectors, using time-varying VAR connectedness on daily returns and volatilities, 2018-2026, and a point-in-time constituent panel
- A 1-point rise in a sector's top-3 share cuts its net shock transmission by 0.47 points ($t = -3.6$); showed that 90 of 110 apparent Granger lead-lag links are artifacts of GARCH errors and vanish under robust tests

RateWalk: Forecasting Fed Decisions with a Shrunk Markov Chain [Paper](#) · [Code](#)

Working paper, 2026

- Forecast the next Fed rate decision from a Markov chain on policy moves and inflation regimes, reading CPI only as first published so revisions never leak; validated out of sample on the US, UK, and Germany, 1990-2026
- Regime conditioning overfit; empirical-Bayes shrinkage fixed it in all three countries, and a blend with a Treasury-curve proxy for market pricing beat both (log-loss 1.056 vs 1.199, 72.7% hit rate over 319 decisions)

Volatility-Managed Factor Portfolios: A Replication and Extension [Paper](#) · [Code](#)

SSRN preprint, 2026

- Extended Moreira-Muir (2017) to the Fama-French five factors plus momentum, 1963-2026, sizing each factor by its prior-month realized variance so leverage uses only past data; alpha tested with Newey-West spanning regressions
- Timing pays only on crash-prone factors: momentum alpha 8.9% a year ($t = 5.0$; Sharpe 0.51 to 0.89, 0.81 after 14 bp per-turn costs) and profitability 2.3% ($t = 2.9$); nothing for size, value, or investment

VolEdge: Measuring the Variance Risk Premium from Option Chains [Paper](#) · [Code](#)

Working paper, 2026

- Built a full-stack platform (Python, FastAPI, React) that extracts risk-neutral variance, skew, and kurtosis from live option chains without a pricing model (Bakshi-Kapadia-Madan) and sets them against realized moments
- Net of costs, 2019-2024: a GARCH volatility-targeted strategy beat SPY on risk (Sharpe 0.85 vs 0.68, 41% smaller drawdown) while naive short-variance harvesting lost money (Sharpe -0.16), so the premium pays for crash risk

Multi-Scale Regime Detection for Adaptive Sector Portfolios [Paper](#) · [Code](#)

SSRN preprint, 2026

- Four-signal crash detector (CUSUM, correlation, breadth, skew) that de-risks only when two agree: 2009-2025 max drawdown -6.2% vs -41.7% for SPY (Sharpe 0.97 vs 0.58); unchanged, Sharpe 1.21 on factor and 1.04 on international portfolios

Also built: [ChronoFund](#), point-in-time SEC EDGAR/XBRL engine (65 tests); [PinSight](#), ODTE risk-neutral densities, live paper trader

EXPERIENCE

New York University, Tandon School of Engineering

Course Assistant, FRE-GY 6103 Valuation for Financial Engineering (Prof. David Shimko)

Contributor, Valuation Principles, Prof. Shimko's graduate valuation textbook (Wiley, in preparation)

Designated lead of four course assistants; resolve questions on mortgage-pool prepayment and Treasury curve bootstrapping

Restructured the textbook's Monte Carlo and variance-reduction material and built its figures, companion Python notebooks, Excel models, and a 164-page solutions manual; chapters serve as the course text

New York, NY

Sep 2026 - Present

Jun 2026 - Present

SKILLS

Programming: Python (pandas, NumPy, SciPy, statsmodels, scikit-learn), SQL, R, C++ (coursework); FastAPI, React, Git, LaTeX

Data and training: Bloomberg Terminal, WRDS, SEC EDGAR/XBRL, FRED/ALFRED; ARPM Quant Bootcamp (2026)